

Multiplicity: Type-Theoretic Knowledge Representation — Blueprint

Core Objects and Laws

Prime-labeled carriers. Elements carry unique primes. A state is (X, H) with lattice $T_n = \mathbb{Z}/n\mathbb{Z}$, signal space $X = \{x : T_n \rightarrow \mathbb{R}^k\}$, and hypergraph $H = (V, E, \iota)$.

Operator words. For each prime p and level $r \geq 1$ define families acting on states and relations:

- Subdivision S_p , rotation $R_\phi^{(p^r)}$, accent $A_\alpha^{(p^r)}$, permutation $W_\pi^{(p)}$.
- Relation operators $\widehat{\text{split}}_p, \widehat{\text{merge}}_p, \widehat{\text{fold}}_p, \widehat{\text{rel}}_p$.

A **prime word** $\widehat{\mathcal{P}}(p)$ composes these; **composite word** multiplies over all primes dividing n . Noncommutation is essential: order matters.

Projectors. Level projectors $\Pi_{p^r} : X \rightarrow X$ and spikes Δ_{p^r} pick invariant and event content at level p^r . Idempotence: $\Pi_{p^r}^2 = \Pi_{p^r}$. Bounds: $\|\Pi_{p^r}\|_2 \leq 1$.

Multiplicity matrix. Time-varying spectral carrier $M(t) = \sum -ip^{m_i(t)} v_i(t) v_i(t)^\top$, $\dot{m}_i(t) = f_i(m(t), \text{couplings})$.

Resonance functional. Validation map $R(H, \{\hat{O}\}; D) \in [0, 1]$ comparing generated structure to data D . Proof-by-resonance: choose operator words maximizing R .

1. Dependent Type Theory (DTT)

1.1 Universes and indices

- Universe of primes $\mathcal{U}_p$. Type of prime labels: $\text{Prime} : \text{Type}$ with decision procedures.
- Type family for multiplicity-indexed carriers: $\text{Carrier} : (p:\text{Prime}) \rightarrow (r:\mathbb{N}) \rightarrow \text{Type}$.
- Time lattice as an indexed type: $T : (n:\mathbb{N}) \rightarrow \text{Type}$ with CRT structure.

1.2 Operators as dependent functions

- $S : (p:\text{Prime}) \rightarrow \text{Carrier } p \ r \rightarrow \text{Carrier } p \ (r+1)$.
- $\Pi : (p:\text{Prime}) \rightarrow (r:\mathbb{N}) \rightarrow X \ n \rightarrow X \ n$ with proofs $\Pi \ p \ r \circ \Pi \ p \ r = \Pi \ p \ r$.
- Noncommutation witnesses: $\neg\text{comm} : \Sigma (p \ q \ r)$. Rewrites $(W \ q \circ A \ p \ r) \neq (A \ p \ r \circ W \ q)$.

1.3 Refinement and constraints

- Refinement type for invariants: $\text{Inv } X := \{x:X \mid E[x]=E_0 \wedge F[x] \geq F_0\}$.
- Effect typing for relation-state coupling using indexed monads $\text{Rel } p : M \ X$ to encode $\widehat{\text{split}}$, $\widehat{\text{merge}}$.

1.4 Proof obligations

- Soundness of Π : subject-reduction on indices $p^{\wedge r} \mid n$.
- Norm bounds: $\|\Pi\|_2 \leq 1$. Provide constructive proofs via averaging definitions.
- Confluence for admissible rewrites inside a resonance class $[W]_R$.

2. Higher-Order Logic (HOL)

2.1 Signatures

- Sorts: $\text{Prime, Level, Tick, Vertex, Edge}$.
- Functions: $\text{divides} : \text{Level} \times n \rightarrow \text{Bool}$, $\text{Pi} : \text{Level} \rightarrow X \rightarrow X$, $\text{Delta} : \text{Level} \rightarrow X \rightarrow X$.
- Predicates: $\text{Admissible}(W,n)$, $\text{PreserveInv}(W,X,H)$.

2.2 Axioms and theorems

- Idempotence, projector-rotation commutation at same level: $\text{Pi } \ell \circ R^{\wedge\{(\ell)\}} = \text{Pi } \ell$.
- Cross-prime noncommutation: existence of $W\pi$ with $[\text{Pi}_{\{p^{\wedge r}\}}, W\pi_q] \neq 0$ for $p \neq q$.
- Completeness for resonance equivalence on finite n : normal forms up to automorphism.

3. Logical Frameworks (LF, CMTT)

3.1 LF signatures

- Kinds: ty , tm . Types: $\text{prime} : \text{ty}$, $\text{lvl} : \text{ty}$, $\text{op} : \text{ty}$.
- Judgments: $\Gamma \vdash W : \text{op}$, $\Gamma \vdash n : \text{tm}$, $\Gamma \vdash \text{admissible } W \ n$.
- Rules: formation/intro/elim for Π, Δ, S, R, A, W , split/merge/fold .

3.2 Contextual modal type theory

- Contexts carry level stacks: $\Gamma \langle p^{\wedge r} \rangle \vdash t$. Modalities $\diamond_{\{p^{\wedge r}\}}$ for lifted action; laws for $\uparrow$ and projection back.

4. Unification Theory

4.1 Equational theory

- Rewrite basis E :
 - $\Pi_{\{p^r\}} \circ \Pi_{\{p^r\}} \rightarrow \Pi_{\{p^r\}}$
 - $\Pi_{\{p^r\}} \circ R^{\{(p^r)\}} \rightarrow \Pi_{\{p^r\}}$
- Cross-prime braid schemata giving critical pairs with finite join only under projector orderings.

4.2 Higher-order pattern unification

- Variables range over operator words subject to `Admissible` and index constraints.
 - Goal: `find W` s.t. `R(W[x];D) ≥ τ` with constraints resolved by ELPI-style HO pattern unification.
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5. Programming Languages and Systems

5.1 Lambda Prolog / ELPI

- Predicates: `op_word(W)`, `admissible(W,n)`, `apply(W, X, X')`, `resonance(W, D, R)`.
- Constraint solving: generate-and-test with pruning via projector normal forms.

5.2 Twelf

- Encode judgments `of W op`, totality theorems for evaluator `eval : W -> X -> X'`.
- Meta-theorems: preservation of invariants, progress for admissible words.

5.3 Beluga

- First-class contexts for level stacks; programs that transform `(X,H)` with contextual proofs bundled.

5.4 F*

- `module MOC`: types for `Prime`, `Level`, `Word`. Ghost proofs for `||Π||2≤1`, `PreserveInv`.
- Extraction for real-time systems; SMT lemmas for divisibility and CRT decomposition.

5.5 TypeDB

- Schema-as-types: `Concept[p]`, `Edge[p]` for native n-ary hyperedges.
 - Polymorphic roles with level constraints; integrity rules mirroring `Admissible`.
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6. Knowledge Representation Evolution

6.1 Beyond RDF triples

- Native hypergraph with node and edge attributes; $\text{arity}(e) \geq 2$ unrestricted.
- Mapping layer to/from RDF: lossless only for binary subgraph.

6.2 Ontology verification

- Consistency as satisfiability of invariant-preserving operator words.
- Satisfiability checks via DTT refinements and Twelf totality.

6.3 Context-sensitive knowledge

- Worlds as $\Gamma(\text{context})$ with indices for time, place, perspective.
- Versioned facts carry (p^r, t) labels; queries project with $\Pi_{\{p^r\}}$.

6.4 Constraint integration

- Unified spec and data validation using refinement types; executable checkers in F^* .
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7. Advanced Applications

7.1 AI memory systems

- Store items with prime labels; retrieval uses Π and resonance search.
- Guarantees: sublinear retrieval under sparse level overlap; proofs via projector orthogonality bounds.

7.2 Personal knowledge graphs (edge AI)

- On-device hypergraph; privacy by selective projection $\Pi_{\{p^r\}}$ before sync.
- Auditability: invariants log preserved.

7.3 Hypergraph ontologies

- Native n-ary relations; operator algebra defines schema evolution as words acting on H .

7.4 Scientific knowledge modeling

- Dynamic theories as time-dependent multiplicities $m_i(t)$; falsifiable via shifts in resonance R on new data.
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8. Implementation Strategy

8.1 Minimal toolchain

- Core: ELPI for search; Twelf/Beluga for meta; F* for verified kernels; TypeDB for storage.

8.2 Data model

- Choose base cycle n ; factorization yields prime levels. Persist (X, H) snapshots.

8.3 Algorithms

- Normal-form reduction using projectors; beam search over operator orders; gradient-free resonance ascent.
- Unification-driven constraint solving for Admissible .

8.4 Testing

- Unit: projector laws, noncommutation witnesses, invariant preservation.
- System: end-to-end resonance improvement on time-series and graph datasets.

8.5 Performance

- Precompute $\Pi_{\{p^r\}}$ via FFT-like averages. Cache CRT index maps. Parallelize over primes.

9. Falsifiable Predictions and Measurements

1. **Signal datasets.** Operator-word features yield $\Delta R \geq 0.1$ AUROC lift over additive baselines on periodic anomaly detection.
2. **Graph dynamics.** Hypergraph tasks with n-ary edges show lower structural loss when trained with projector-constrained updates.
3. **Cognitive retrieval.** Prime-level projections reduce interference; measurable drop in retrieval time vs. flat embeddings.

Each prediction includes protocols: datasets, metrics, ablations on operator order.

10. Appendix: Typed Skeletons

- DTT signatures for $\text{Prime}, \text{Level}, \Pi, \Delta, S, R, A, W$.
- ELPI predicates and sample rules.
- F* interfaces with lemma stubs for projector bounds and preservation.